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Distinct carbon sequestration in the glacial Pacific despite vigorous deep ocean circulation

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ABSTRACT

The Pacific Ocean holds the largest marine carbon inventory and serves as a vital sink for atmospheric CO_2 during glacial periods. Associated with drastic cooling of the global oceans, the Pacific carbon sequestration has traditionally been attributed to slow deep circulation, similar to the mechanisms possibly occurring in the glacial Atlantic Ocean. Recently, however, this view has come under scrutiny due to a compilation of neodymium isotope records showing a shorter, rather than longer, transit time for deep Pacific water masses. The noted discrepancy suggests the presence of differing carbon dynamics operating in the glacial Pacific, raising the question of whether a larger carbon reservoir could persist under stronger deep ocean circulation. Here, we employ biogeochemical simulations to examine the possible mechanisms driving the enhanced carbon inventory in the glacial Pacific. The results indicate that the proposed rapid movement of deep Pacific water masses may be linked to cooling in the Southern Ocean, which further increases carbon storage primarily by intensifying air-sea disequilibrium at surface and impeding diapycnal mixing within the ocean interior. Essentially, the results suggest that a substantial Pacific carbon inventory can withstand intensified deep circulation. In the context of thermohaline circulation, the glacial North Atlantic plays a crucial role in absorbing atmospheric CO_2, whereas the surface Southern Ocean and North Pacific operate synergistically to inhibit the release of carbon from the ocean.

1. Introduction

Air bubbles preserved in polar ice cores (Lüthi et al., 2008) record a decline of 80 to 120 ppm in atmospheric pCO_2 (pCO_2^{atm}) from warm interglacial (IG) to cooler glacial (G) states. The global ocean is considered the main reservoir for the carbon lost from the atmosphere during glacial periods, especially as terrestrial biomass likely decreased during these times (Peterson et al., 2014).

The Pacific Ocean, which accounts for around 50 % of the total volume of the global oceans and serves as the most extensive oceanic reservoir for nutrients and dissolved inorganic carbon (DIC), shows a notable increase in carbon storage during glacial periods. For example, proxies reflecting the oxygenation state, such as redox-sensitive uranium (Jacobel et al., 2017) and magnetite dissolution (Korff et al., 2016), demonstrate a distinct reduction in oxygen levels within the glacial deep Pacific. Remineralization of organic carbon consumes ambient oxygen, such that oxygen depletion is tied to a more extensive respiratory carbon reservoir. Furthermore, traditional interpretations based on radiocarbon data (De La Fuente et al., 2017; Lund et al., 2011; Marchitto et al., 2007; Stott et al., 2009) suggest that the ventilation age of Pacific water masses rises during glacial periods, typically supporting greater carbon sequestration in the Pacific.

Although these reconstructions align on the rise in Pacific carbon inventory, there is still contention regarding the mechanisms that may have caused this increase. Based on neodymium isotope (εNd) and radiocarbon data, it has been suggested that, akin to the conditions observed in the glacial deep Atlantic, the rise in carbon storage within the deep Pacific could be attributed to a slower deep circulation during glacial periods, resulting in a prolonged residence time for the deep ocean water masses (Du et al., 2018; Skinner et al., 2015). As the North Pacific (NP) currently serves as the far end of the thermohaline circulation, a rise in residence time consequently facilitates a more thorough remineralization of organic materials, thus enhancing the regenerated carbon inventory.

Nonetheless, another recent compilation of εNd records (Hu and Piotrowski, 2018) in the Pacific has challenged this interpretation by demonstrating that the transit time of water masses significantly decreases during glacial periods. In other words, the deep Pacific could

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have exhibited a more vigorous circulation during glacial periods, rather than a sluggish one (Du et al., 2020; Rafter et al., 2022). While the ongoing debate remains unresolved regarding the deep Pacific circulation mode, the findings by Hu and Piotrowski (2018) raise an unexplored question: is it possible for the deep Pacific to have held a larger carbon pool under a vigorous circulation pattern?

If this case were true, one possible mechanism is the air-sea disequilibrium of CO_2 exchange (e.g., Khatiwala et al., 2019). This disequilibrium is marked by the observed difference in the atmospheric and sea surface pCO_2 that deviates from a state of chemical equilibrium under specific steady conditions (Ito and Follows, 2013). For example, in the high-latitude Southern Ocean (SO), water masses that upwell from the ocean's interior contain elevated levels of CO_2, which causes the surface pCO_2 to exceed atmospheric pCO_2, ultimately leading to outgassing from the ocean. The process of outgassing would lead to a decrease in the surface pCO_2, consequently diminishing the pCO_2 difference (disequilibrium) between the surface water and the atmosphere. In a theoretical framework, this difference would eventually decrease, leading to a state of equilibrium, provided there is sufficient time for the upwelled water and the atmosphere to exchange CO_2 prior to the water masses sinking back into the deep ocean (Khatiwala et al., 2019). Current observations reveal that a state of disequilibrium prevails in the surface SO (Bakker et al., 2016), suggesting that a fraction of CO_2 expected to degas from the SO is instead being sequestered in the ocean. Under glacial conditions, although a number of studies highlight the role of air-sea disequilibrium in lowering glacial pCO_2^{atm} (Khatiwala et al., 2019; Marzocchi and Jansen, 2019; Ödalen et al., 2018), whether it has contributed to the basin-scale carbon sequestration in the Pacific remains inadequately explored.

The mechanisms behind deep Pacific carbon sequestration remain unresolved, and particularly puzzling is the surface NP CO_2 source-and-sink pattern tied to alterations in the deep carbon reservoir. Currently, the subpolar surface NP acts as a carbon source to the atmosphere on an annual basis (Takahashi et al., 2009), due to the upwelling of carbon-rich water from the ocean interior (Nishioka et al., 2020). Conversely, the North Atlantic (NA) absorbs CO_2 from the atmosphere due to relatively low surface pCO_2 (Takahashi et al., 2009). Recent research reveals that the glacial NA exhibited even greater efficiency in carbon absorption driven by the intensified latitudinal sea surface temperature (SST) gradient (Yu et al., 2019). Although the subpolar NP experienced significant cooling as well, the existence of an enhanced solubility pump requires further investigation. On the other hand, boron isotopes, serving as an indicator of surface water pH, reveal diminished pCO_2 difference between the surface NP and the atmosphere during the Last Glacial Maximum (LGM) compared to the Holocene (Gray et al., 2018). This likely suggests a decline in the NP surface carbon exchange rates. Under this substantial cooling, the glacial NP is therefore associated with two opposing processes affecting the surface CO_2 flux: substantial cooling that may promote carbon uptake flux, similar to the glacial North Atlantic, and diminished surface pCO_2 difference that may reduce carbon exchange.

Overall, the uncertainties surrounding deep carbon sequestration and surface carbon exchange in the glacial Pacific raise a fundamental question: did the basin-scale carbon dynamics in the Pacific exhibit patterns similar to those observed in the Atlantic? To address this, we performed and analyzed a collection of biogeochemical model outputs, clearly demonstrating that the glacial Pacific could act as a major carbon reservoir in spite of vigorous deep ocean circulation during glacial periods, which could be systematically distinct from an Atlantic-like mode.

2. Model and experimentation

This study aims to enhance our understanding of how alterations in glacial ocean conditions may contribute to increased carbon storage in the Pacific, presumably tied to more vigorous deep ocean circulation. To this end, we utilized a cutting-edge ocean biogeochemical model, specifically a coupled version of the Massachusetts Institute of Technology ocean general circulation model (MITgcm) (Campin et al., 2020), coupled with the Regulated Ecosystem Model 2 (REcoM2) (Hauck et al., 2013). The physical setup of the MITgcm extends from 80°S to 80°N in latitude, coupled with a dynamic-thermodynamic sea ice model (Losch et al., 2010). The longitudinal resolution is approximately 2°, whereas the latitudinal resolution varies between 0.38° and 2° The model features a maximum depth of 5500 m, comprising 30 vertical layers that are unevenly distributed.

The biogeochemical model REcoM2 effectively simulates the cycles of carbon, nitrogen, iron, and silicon in the ocean, incorporating flexible intracellular stoichiometry across phytoplankton, zooplankton, and detritus. A dynamic and globally uniform atmospheric pCO_2 box, as outlined by (Du et al., 2022), was integrated into REcoM2. This integration ensures that the total carbon content within the ocean and atmosphere system remains constant across all simulations, facilitating a more thorough investigation into variations in basin-scale carbon reservoirs. The MITgcm-REcoM2 was driven by atmospheric forcing obtained from a coupled climate model study (Zhang et al., 2013). For a detailed description of forcing processing and experiment initialization, please refer to the Texts S1 and S2 in the Supplementary Materials.

This research involved the analysis of four experiments, namely IG_ctl, IG_Gap, IG_Gsap, and G_ctl. In particular, IG_ctl and G_ctl were derived from pre-industrial (PI) and LGM conditions, exemplifying typical interglacial and glacial states. For better readability, IG_ctl and G_ctl are referred to as the 'Interglacial Experiment' and the 'Glacial Experiment' in the following discussion, respectively. Admittedly, each interglacial period may differ from the others in the orbital configuration and consequently insolation (Tzedakis et al., 2009). Nonetheless, considering that the IG_ctl aligned with the PI conditions was well calibrated with available observations previously (Pagnone et al., 2019), we take our analysis as a comparison between IG and G conditions for a generic understanding of process, rather than a transient simulation of the Pacific carbon inventory during a specific glacial cycle.

To reveal mechanisms driving the Pacific carbon sequestration under possible vigorous circulation, we utilized alterations in SSTs as a surrogate during the experimental design. A recent compilation shows changes in SST in excess of −8 °C in the NA, NP, and the Pacific sector of the SO during the LGM (Tierney et al., 2020). As SST serves as a vital connection between the atmosphere and ocean, the documented decline in SST during glacial times would impact various processes, including sea ice formation (Marzocchi and Jansen, 2019), buoyancy forcing (Watson et al., 2015), and carbon exchange rates (Hauck et al., 2013). Crucially, previous studies (e.g., (Marzocchi and Jansen, 2019, 2017)) have revealed a pivotal role of temperature in driving changes in oceanography, featuring reorganization of the water mass geometry and denser production of the Antarctic Bottom Water (AABW). Further, our previous research, together with the above model studies, has confirmed changes in the SST can lead to substantial alterations of circulation in both the Atlantic and Pacific basins (Du et al., 2022), specifically in the physical model MITgcm.

In practice, alteration of the SST represents the result of deliberate modification of glacial-interglacial atmospheric forcing (please see the 'Processing of forcing fields' section in the Supplementary Materials), and leads to specific deep circulation patterns as will be discussed. In this regard, two cooling experiments, IG_Gap and IG_Gsap, incorporating cooling in designated ocean regions (Figure S1), are performed and analyzed. Specifically, in IG_Gap (referred to as the 'Northern Cooling Experiment'), the surface NP and NA (north of 30 °N) experienced a cooling transition of roughly 6 to 8 °C. This experiment aims to evaluate whether the pronounced oceanic cooling in the Northern Hemisphere can lead to the hypothesized vigorous deep Pacific circulation, and further substantial carbon accumulation in the Pacific. In IG_Gsap (referred to as the 'Bi-polar Cooling Experiment'), cooling in IG_Gap was further combined with cooling in the SO (defined as south of 45 °S, with a magnitude of roughly 3.5 °C), leading to a significant

cooling observed in high-latitude oceans across both hemispheres. A comparison between IG_Gsap and IG_Gap would enhance our understanding of the potential interaction between the bipolar processes, which are believed to impact the Pacific carbon cycle during the glacials, either synergistically or antagonistically. Specifically, the comparison between the two cooling experiments could possibly reveal the impact of air-sea disequilibrium on the Pacific carbon storage, as the degree of disequilibrium can be largely influenced by the SSTs (Marzocchi and Jansen, 2019).

We would like to note that within the set of experiments, IG_Gap (Northern Cooling Experiment) refers to the new experiment in the current study, and IG_Gsap (Bi-polar Cooling Experiment), previously published but unanalyzed, is examined in the current study. The remaining two (IG_ctl and G_ctl) were analyzed with an emphasis on the Atlantic Ocean (Du et al., 2022).

3. Results and discussion

3.1. Modeled glacial physics and carbon cycle

The Glacial Experiment (G_ctl), serving as a depiction of typical glacial conditions, demonstrates a strong alignment regarding the SST, sea ice cover, and carbon cycle with prior research. The modeled mean glacial SST is 15.1 °C, representing a decrease of 2.9 °C compared to the Interglacial Experiment (IG_ctl). The observed anomaly corresponds with the output of the coupled model (2.8 °C) from which the forcing fields were obtained (Zhang et al., 2013) and falls within the estimated range of SST differences (2 to 4 °C) reported in other model-based studies (Alder and Hostetler, 2015; Buchanan et al., 2016). Significant cooling is noted in the NA, NP, and SO (Fig. 1a), corroborating the proxy-based reconstruction presented by (Tierney et al., 2020). It is important to note that in the cooling sensitivity experiments, the cooling of a designated area (NP, NA, and/or SO) leads to a comparable regional pattern in SST (Figure S1) and sea ice concentration.

The extent of sea ice coverage surrounding Antarctica significantly increases under glacial conditions (Fig. 1b). The peak extent of sea ice in the Southern Hemisphere in G_ctl reaches 31.1 $\times$ 10^6 km^2, which is just over double the extent observed in IG_ctl at 15.2 $\times$ 10^6 km^2. Notably, even in the summer months of the Southern Hemisphere, the Glacial Experiment retains an ice-covered area of 17.8 $\times$ 10^6 km^2, compared to a mere 3.5 $\times$ 10^6 km^2 under interglacial conditions. The experiments demonstrate an adequate representation of Southern Hemisphere sea ice cover in relation to reconstructions during interglacial periods (Gloersen et al., 1993); however, the Glacial Experiment tends to slightly underestimate the peak Southern Hemisphere sea ice cover when compared to the reconstruction by (Gersonde et al., 2005). As the SO is cooled to the glacial levels in the Bi-polar Cooling Experiment (IG_Gsap), there is a significant expansion of sea ice, mirroring the trends observed under glacial conditions.

The calibrated marine biogeochemical model exhibits strong agreement with observational data (refer to (Pagnone et al., 2019)), with pCO_2^{atm} reaching equilibrium at 276 ppm. Under glacial conditions (G_ctl), the modeled pCO_2^{atm} is 223 ppm, representing a decrease of 53 ppm relative to interglacial conditions. This significant reduction in pCO_2^{atm} accounts for over half of the recorded ~90 ppm decline in CO_2 levels during glacial periods (Lüthi et al., 2008). Nonetheless, it is possible that the model has not fully captured the extent of this drawdown, due to its simplified representation of marine sedimentation processes. The glacial ocean holds around 112 gigatons (Gt, 1 Gt = 10^{15} g) more DIC compared to the interglacial ocean, with about two-thirds of this extra carbon residing in the deep abyssal ocean beneath 2000 m (Table 1). At the basin scale, the deep Pacific demonstrates a significant rise in DIC storage, amounting to 28 Gt. As reported previously (Du et al., 2022), this carbon storage pattern exhibits a strong correlation with contemporary modeling research (Buchanan et al., 2016; Yamamoto et al., 2019). In the subsequent sections, we utilize the Glacial Experiment as a reference point to explore the surface forcing influencing the carbon dynamics in the Pacific region.

3.2. No substantial pacific carbon sequestration under the northern hemisphere cooling

The cooling experiments (IG_Gap and IG_Gsap), initialized from interglacial states (IG_ctl) with cooling in designated regions, serve to investigate the potential drivers for the increased Pacific carbon storage and changes in the surface carbon exchange. Particularly, as both the NA and the NP experienced an evident cooling of over 8 °C at the glacial time, we first take the Northern Cooling Experiment (IG_Gap), in which the subpolar NA and NP were cooled to G states, to examine the Pacific carbon cycle response to the cooling of the Northern Hemisphere oceans.

Contrary to expectations, there is little Pacific carbon sequestration in this experiment. Instead, the modeled pCO_2^{atm} rises by 9 ppm relative to the interglacial conditions (Table 1), which is tied to the release of carbon from the deep Pacific. The overall decline in carbon storage within the Pacific amounts to 29 Gt C (Table 1), with the deep Pacific, specifically below 2000 m, experiencing a loss of 45 Gt of carbon. The greater carbon loss observed in the deep Pacific compared to the entire basin suggests an elevated carbon inventory in other regions of the Pacific. Fig. 2b illustrates the zonal mean anomaly in DIC concentration. The thick black contour marks where the DIC anomaly = 0, distinctly separating the Pacific into two regions: a noticeable carbon loss in the Pacific Deep Water located in the deep NP; and a significant carbon gain

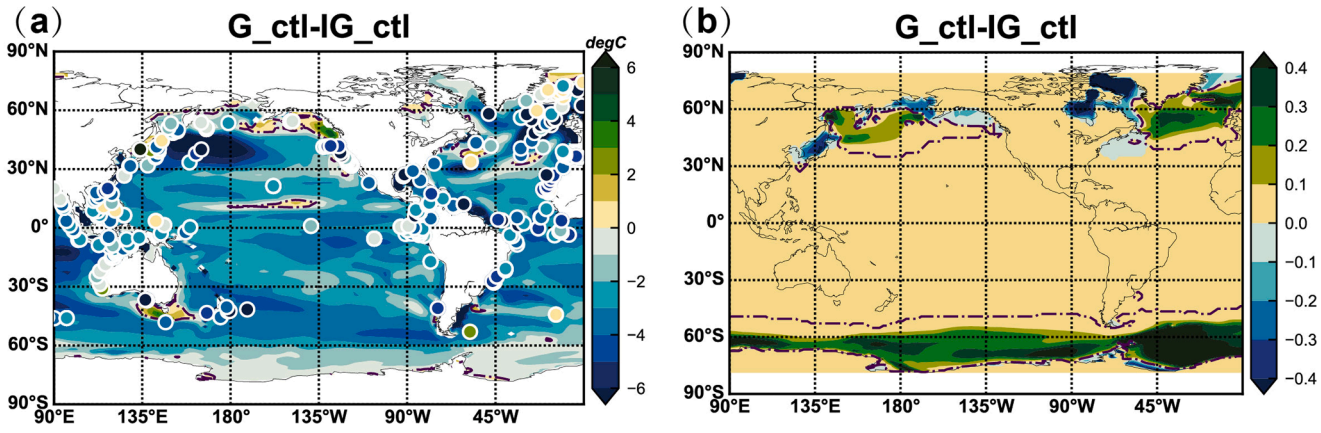


Fig. 1. Anomaly of modeled SST (a) and sea ice concentration (b) between glacial (G_ctl) and interglacial (IG_ctl) conditions. In (a), imposed scatter points are associated with proxy-based reconstruction utilizing calibrated SST anomalies from the Last Glacial Maximum to the Late Holocene, as derived from a recent compilation (Tierney et al., 2020).

Table 1
Overview of experimental design and changes in carbon reservoirs.

Exp.name	SO Cooling	NA Cooling	NP Cooling	ΔpCO_2^{atm} (ppm)	ΔC_g (Gt)	$\Delta C_{g\text{-}b2k}$ (Gt)	ΔC_p (Gt)	$\Delta C_{p\text{-}b2k}$ (Gt)
IG_ctl (Interglacial)	x	x	x	–	–	–	–	–
IG_Gap (Northern Cooling)	x	✓	✓	+9	−19	−67	−29	−45
IG_Gsap (Bi-polar Cooling)	✓	✓	✓	−32	+68	+40	+20	+11
G_ctl (Glacial)	✓	✓	✓	−53	+112	+75	+50	+28

Table 1. Overview of experimental design and changes in carbon reservoirs. In Table 1, "SO", "NA", and "NP" refer to the Southern Ocean, the North Atlantic, and the North Pacific, respectively. In the "Cooling" columns, "✓" means the specific region(s) are cooled to the LGM state, while "x" means the region(s) remains at the interglacial (PI) temperature. In the "ΔpCO_2^{atm}" column, the pCO_2^{atm} difference between the given experiment and IG_ctl is shown. Columns with "ΔC" indicate the changes in the basin-scale carbon reservoirs; the footnotes "g" and "p" indicate the global ocean, and the Pacific Basin, respectively; the footnote "b2k" refers to "below 2000 m".

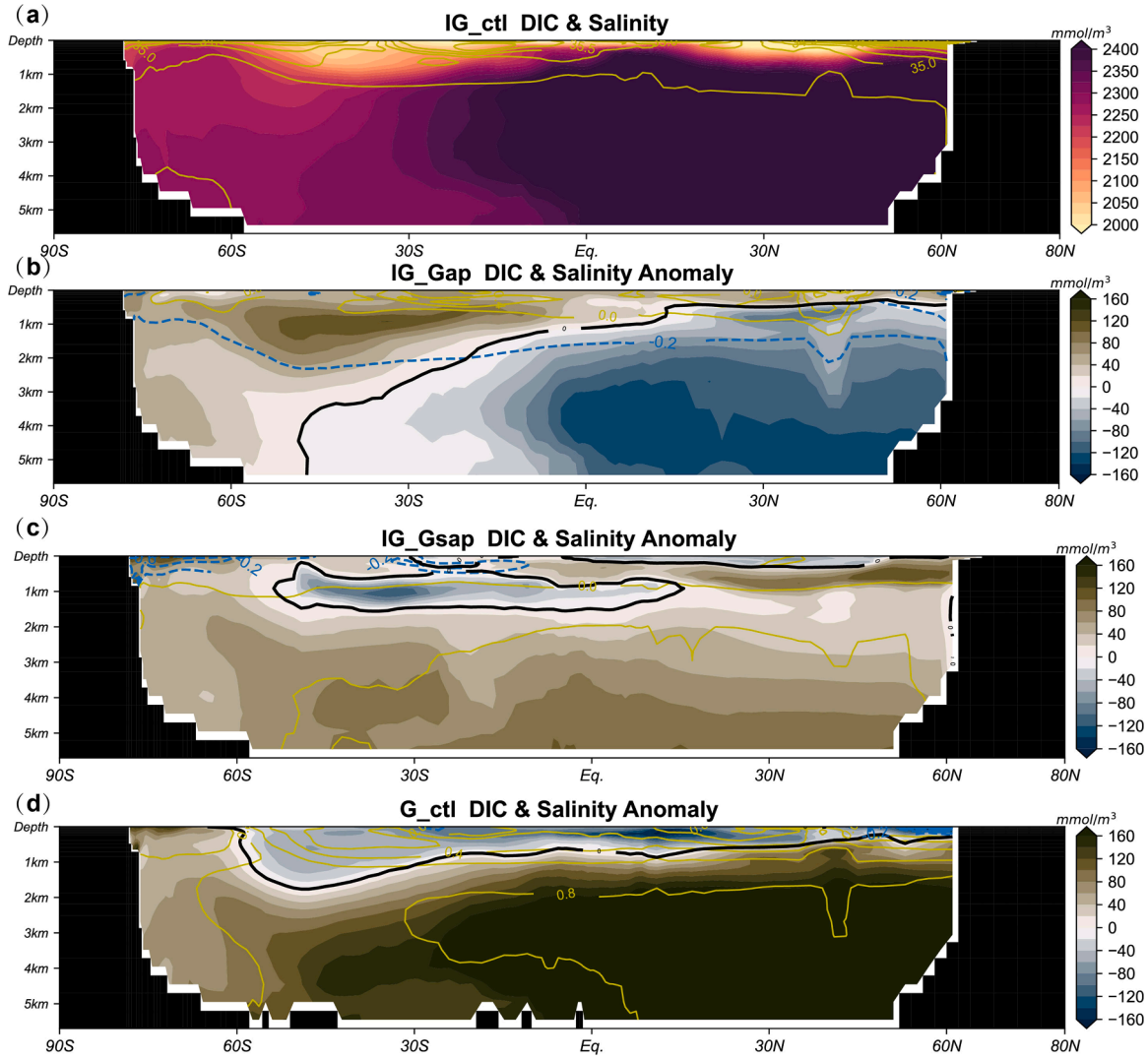


Fig. 2. Zonal averaged DIC concentration (shading) and salinity (contours, in g/kg) in the Pacific Basin. Panel (a) utilizes the modeled values for plotting, whereas panel (b) illustrates the anomaly between the Northern Cooling Experiment (IG_Gap) and Interglacial Experiment (IG_ctl). To elucidate the Pacific carbon storage response to the bi-polar forcing interactions, anomalies between the selected experiment and IG_Gap are illustrated in (c) and (d), respectively.

observed in the SO sector, extending up to the surface NP.

The notable positive DIC anomaly observed in the SO sector is primarily driven by a strong teleconnection within the Meridional Overturning Circulation (MOC). Under the cooling of NA, there is an increased production of North Atlantic Deep Water (NADW) from the subpolar region prior to its descent into the ocean's interior, forming a vigorous Atlantic MOC (AMOC, Figure S2a). The strengthened AMOC subsequently results in a general acceleration of the thermohaline circulation, leading to a vigorous mode of the overturning circulation in the deep Pacific (Fig. 3) and the upwelling at the SO (Figure S2b). The

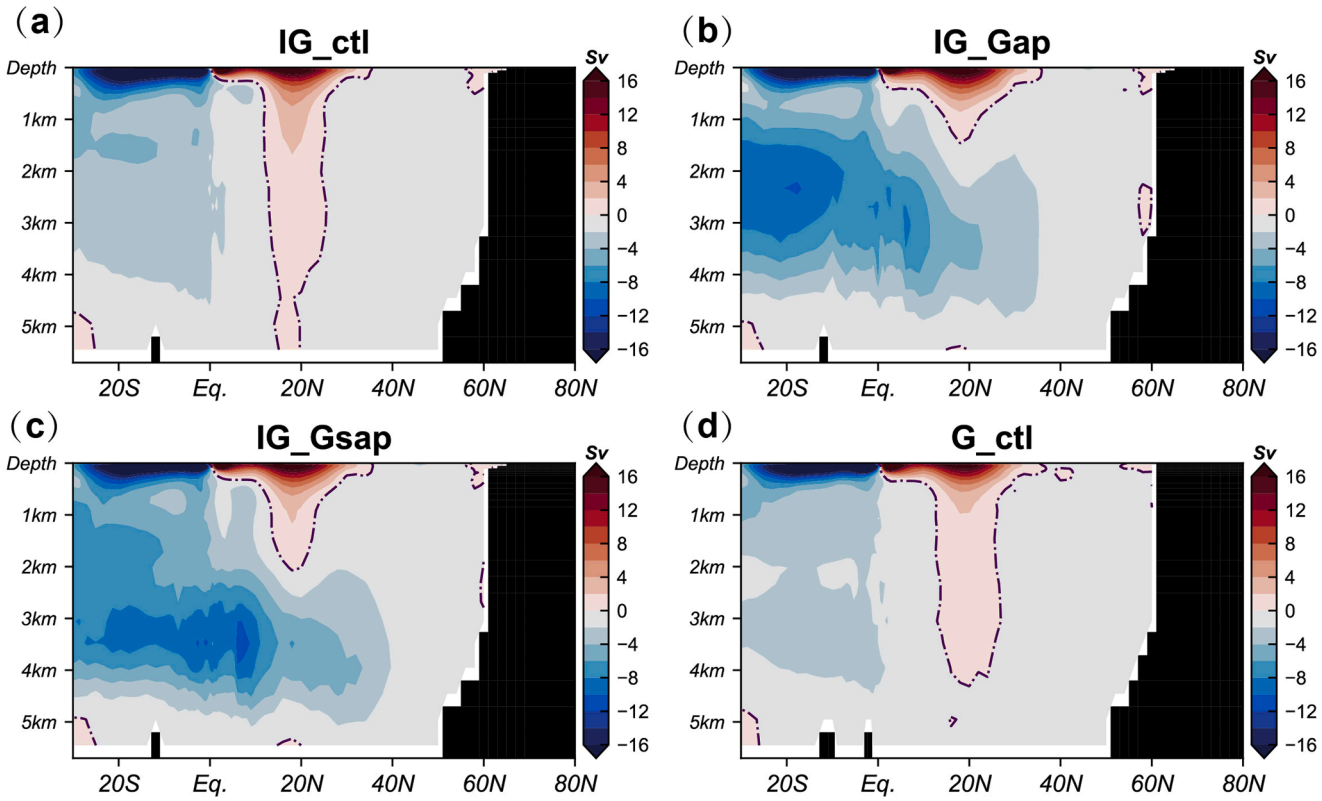


Fig. 3. Modeled zonally integrated meridional stream function (in Sv, 1 Sv = 10^6 m^3/s) in the Pacific basin: (a) IG_ctl, (b) IG_Gap, (c) IG_Gsap, (d) G_ctl. Note that the Bi-polar Cooling Experiment (IG_Gsap) exhibits a vigorous deep Pacific overturning circulation associated with higher DIC concentrations in Fig. 2.

upwelling at the SO plays a vital role in transporting carbon-rich deep water from the abyss to the surface ocean (Yu et al., 2020). Compared to the interglacial state, the SO upwelling is clearly intensified in IG_Gap, functioning to efficiently transfer carbon from the deep NP to the SO, thus influencing the DIC anomaly pattern. Moreover, the increased pace of the thermohaline circulation would result in reduced transit time for the deep Pacific water masses, consequently leading to diminished remineralization of organic matter and promoting carbon loss at the far end of the circulation, specifically in the deep NP.

The positive DIC anomaly observed in the upper NP indicates the influence of regional processes, such as deepening of the mixed layer depth (MLD, Fig. 4) and enhanced solubility pump. Currently, there is no deep convection at the NP (Figure S3). In spite of the low temperature, this phenomenon is primarily driven by the reduced salinity presenting at the subpolar NP (Menviel et al., 2014), resulting in a pronounced halocline and thus stratification. In IG_Gap, however, significant cooling in the surface NP leads to an increase in the annual mean MLD (Fig. 4a), suggesting a perturbation of the NP halocline and stratification. Although this perturbation does not result in a basin-wide deep water formation or expansion, it undeniably affects the regional isopycnals, causing carbon to leak from the deep interior to the surface, as illustrated by the vertical gradient of the DIC anomaly in Fig. 2b Another mechanism that plays a role in the increase of surface DIC is the solubility pump, which is affected by cooling processes. The solubility of CO_2 in seawater is significantly influenced by temperature, as cooler water can retain a greater amount of CO_2 at equilibrium. The significant cooling occurring in the NP is expected to lead to an increased dissolution of carbon in seawater.

The cooling in the Northern Hemisphere subsequently results in a reorganization of surface CO_2 fluxes. As illustrated in Fig. 5a, the CO_2 flux demonstrates increased carbon absorption in the high-latitude NA and NP, while there is a substantial rise in outgassing in the SO. The temperature-driven solubility pump in the NA has been reported previously (Yu et al., 2019), and is confirmed by models (Du et al., 2022). Nonetheless, this greater carbon uptake at the subpolar NP appears to contradict proxy-based reconstructions and the Glacial Experiment

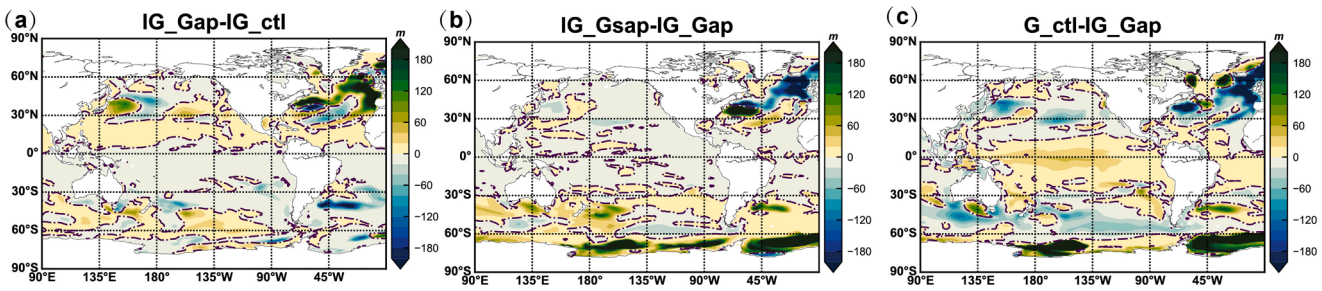


Fig. 4. Modeled annual mean mixed layer depth (MLD) anomalies. In (a), the anomaly between the Northern Cooling Experiment (IG_Gap) and Interglacial Experiment (IG_ctl) is plotted. In (b) and (c), the anomalies are calculated between the Bi-polar Cooling Experiment (IG_Gsap) and the Glacial Experiment (G_ctl) relative to IG_Gap, respectively.

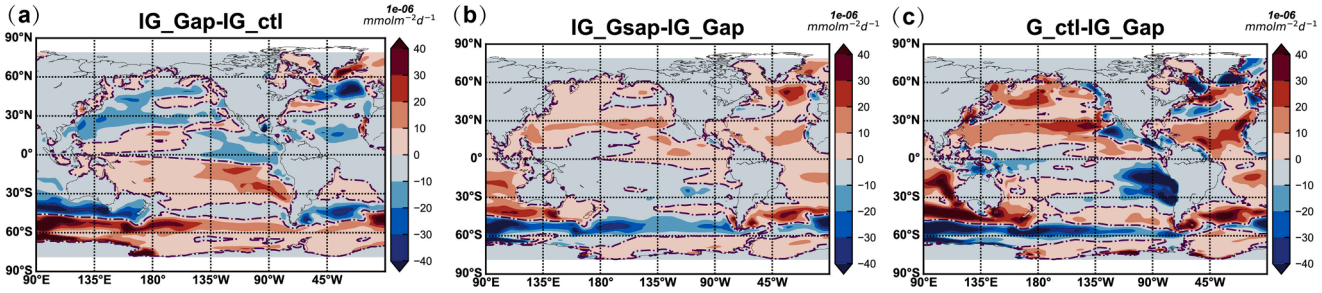


Fig. 5. Surface CO_2 flux anomalies in the experiments. In the plots, positive values refer to enhanced ocean outgassing or reduced ocean uptake. Plot (a) depicts the anomaly between the Northern Cooling Experiment (IG_Gap) and the Interglacial Experiment (IG_ctl). To elaborate on the response of the Pacific carbon storage to the interaction between the bipolar forcing, anomalies between the selected experiment and IG_Gap are plotted in (b) and (c), respectively.

(Figure S4 and Gray et al., 2018), both of which show a decline in the sea-air pCO_2 difference, suggesting suppressed air-sea CO_2 fluxes during peak glacial periods. Moreover, the increase in the surface NP carbon uptake does not lead to an increased ocean carbon storage, as the significant outgassing in the SO offsets the influx of CO_2 in the Northern Hemisphere (Fig. 5a). Consequently, the cooling of the NA and NP yields an increase of 9 ppm in the modeled pCO_2^{atm}.

Overall, these findings in the Northern Cooling Experiment (IG_Gap) eliminate the SST drop in the Northern Hemisphere as the primary control of the glacial Pacific carbon sequestration, despite the fact that the sharp cooling was clearly recorded in the geological records, and vigorous circulation is observed in our simulations (Fig. 3). However, associated with this vigorous deep Pacific circulation, this experiment offers a unique and thought-provoking insight into the carbon cycle of the glacial Pacific. In contrast to the anticipated lower pCO_2^{atm} levels, the larger carbon reservoir in the deep Pacific, the diminished sea-air pCO_2 difference and thus the constrained air-sea CO_2 exchange, and the stable stratification at the NP, the IG_Gap appears to function in a fundamentally different way: exhibiting higher pCO_2^{atm} levels (Table 1), carbon loss from the deep Pacific (Fig. 2b), enhanced sea-air pCO_2 difference (Figure S4), and increased magnitude of surface CO_2 flux (Fig. 5a), all of which may be tied to the disturbances in the stratification at the NP (Fig. 2b and Fig. 4a). These results point to a conclusion that additional processes must negate or even exceed the effects of the perturbed stratification to facilitate the carbon accumulation in the deep Pacific, as will be examined in the analysis of the Bi-polar Cooling Experiment (IG_Gsap).

3.3. Enhanced deep pacific carbon storage in spite of vigorous circulation

The IG_Gsap experiment features a unique scenario, namely, enhanced deep Pacific carbon storage despite vigorous deep ocean circulation. In the experiment, cooling of the NA and NP (as in IG_Gap) was further combined with cooling in the SO. The simulated pCO_2^{atm} is 244 ppm (Table 1), which is 32 ppm lower than that under interglacial conditions and 41 ppm lower than that in IG_Gap. In comparison to IG_Gap, the Pacific reservoir in this experiment has increased by 49 Gt C, while the deep Pacific, situated below 2000 m, holds an additional 57 Gt of carbon. The changes in carbon inventories indicate a significant response of ocean dynamics, and consequently the carbon cycle, to the concurrent cooling in both hemispheres. More importantly, with stronger circulation in the deep Pacific (Fig. 3c), this experiment confirms that a larger carbon inventory (Table 1) in the Pacific can endure despite vigorous circulation, highlighting a distinct contrast with the enhanced carbon inventory in the Atlantic, which was proposed to be influenced by sluggish circulation.

Fig. 2c and 2d illustrate the DIC anomaly of the Bi-polar Cooling Experiment (IG_Gsap) and the Glacial Experiment (G_ctl) in relation to the Northern Cooling Experiment (IG_Gap). It is clear that in both IG_Gsap and G_ctl, elevated DIC concentrations are observed in the SO sector and deep NP, while negative DIC anomalies are limited to shallower depths. The observed pattern in DIC concentration aligns closely with the variations in salinity. Both experiments demonstrate an increase in salinity with depth in the ocean interior. The similarity suggests that IG_Gsap may capture essential mechanisms working to store carbon in the deep Pacific, since G_ctl is a typical glacial scenario.

Our analysis underscores the significance of increased SO air-sea disequilibrium and suppressed diapycnal mixing, both influenced by the cooling in the SO, in the accumulation of the deep Pacific carbon inventory. Previous model studies have shown that, the extent of pCO_2^{atm} influenced by the disequilibrium differs across models (Khatiwala et al., 2019; Marzocchi and Jansen, 2019; Ödalen et al., 2018); however, the consensus indicates that it played a significant role in the reduction of glacial pCO_2^{atm}, particularly during the glacial maxima. Ultimately, this disequilibrium is governed by the duration of surface transit time for a water mass (i.e., the interval between its upwelling and descent) (Stein et al., 2020), which is further influenced by the southward meridional velocity and the extent of sea ice cover in the SO. An increase in either factor would hinder the CO_2 exchange at the air-sea interface, thereby exacerbating the disequilibrium. In our case, following the methodology established by (Stein et al., 2020) (Text S3), we present the surface water transit time is 315 days under interglacial (IG_ctl) conditions (Du et al., 2022). Cooling of NA and NP (as in IG_Gap) leads to a slight increase of the transit time to 342 days. In contrast, when the SO is further cooled (as in IG_Gsap), the transit time drastically drops to 46 days, which is relatively close to the glacial conditions (G_ctl, 26 days, Du et al., 2022). Given that the process of CO_2 reaching equilibrium between the atmosphere and the surface typically spans about a year (Ito and Follows, 2013), sole cooling of NA and NP exhibits consequently a minimal level of disequilibrium (Figure S4) via an increase in the surface transit time, ultimately contributing to the significant outflux of CO_2 from the SO (Fig. 5a). Conversely, both a rising overturning rate (Fig. 3) and an increase in sea ice (Fig. 1b and Fig. 6b), both evident in IG_Gsap and G_ctl, lead to a greater level of disequilibrium, which enhances carbon sequestration in the deep Pacific.

The increase in the disequilibrium alone could not account for the change under the bi-polar cooling (IG_Gsap) and glacial conditions (G_ctl). The Pacific carbon loss noted in IG_Gap is tied to a positive DIC anomaly in both the SO and the upper NP. As discussed previously, the increase in carbon observed in the upper NP can be attributed to a reduction, or potentially a disruption, in stratification, resulting in deep carbon leakage through diapycnal mixing. If this situation persists in IG_Gsap, an enhanced vertical gradient of DIC concentration driven by the air-sea disequilibrium discussed above would consequently lead to more significant carbon leakage in the NP. Nevertheless, IG_Gsap demonstrates a loss of carbon in the upper subsurface NP, suggesting that the vertical mixing of DIC is impeded. We therefore propose an additional factor influencing deep Pacific carbon storage, namely, the SO-driven enhancement of stratification. Cooling of the bi-polar oceans in IG_Gsap encompasses a significantly greater expanse of sea ice in the SO

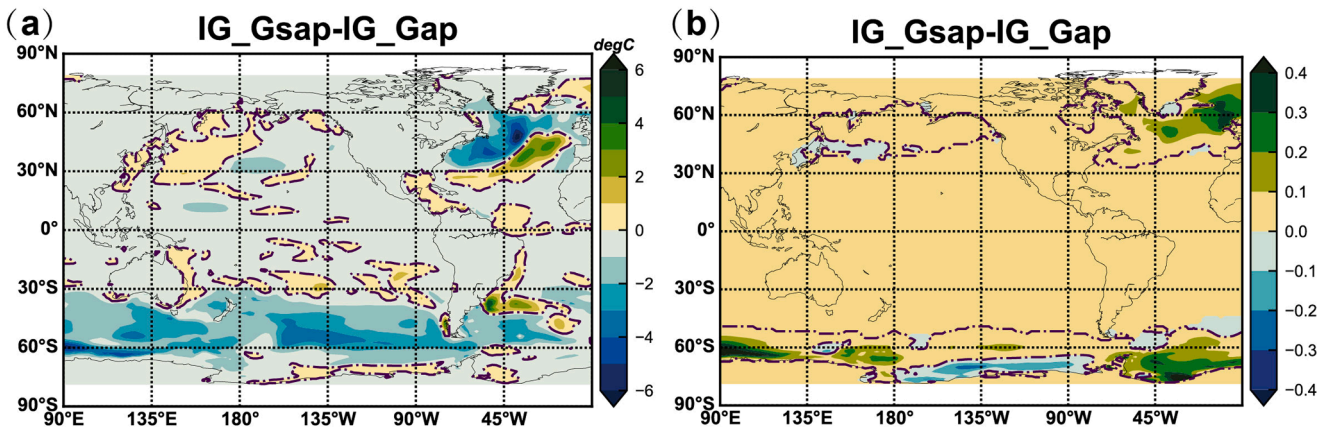


Fig. 6. Modeled SST (a) and sea ice concentration (b) anomalies between the Bi-polar Cooling Experiment (IG_Gsap) and the Northern Cooling Experiment (IG_Gap).

because of cooling in the SO (Fig. 6b). As such, the AABW would become denser during its formation, resulting in lower temperatures and higher salinity due to brine rejection during sea ice formation (Zhang et al., 2013). As the water masses in the Pacific can only ventilate at the SO (Talley, 2013), the denser AABW suggests a more stable density structure, characterized by increased stratification within the ocean's interior. In this scenario, the geochemical tracers would face greater challenges in mixing between isopycnals, thereby hindering carbon leakage from the deep overturning cell. The reduced DIC concentration at the surface Pacific leads to a decrease in the subpolar NP sea-air pCO_2 difference (Figure S4). As a result, we observe positive CO_2 flux anomalies in both IG_Gsap and G_ctl compared to IG_Gap (Fig. 5b, and 5c), indicating that the air-sea flux is substantially restricted, consistent with the interpretation from boron isotope reconstruction (Gray et al., 2018).

Overall, we find that substantial carbon accumulates in the deep Pacific when the high-latitude oceans (NA, NP, and SO) are simultaneously cooled, closely resembling the carbon sequestration pattern observed under glacial conditions (G_ctl). Particularly, IG_Gsap shows that a larger Pacific carbon reservoir can persist in spite of vigorous circulation. Despite the unfavorable shorter transit time of the deep water mass for an expanded regenerated carbon pool, the simulated increased air-sea disequilibrium and diminished diapycnal mixing collaboratively enhance carbon sequestration in the Pacific, counteracting and exceeding the impacts of the vigorous circulation. These processes suggest that the glacial Pacific may have operated systematically differently from the Atlantic in terms of the basin-scale carbon sequestration, as will be discussed in the synthesis section.

4. Synthesis: a comparison between the pacific and atlantic on basin-scale carbon sequestration

Our model results, in conjunction with prior data, point to an unexplored carbon sequestration pattern in the glacial Pacific compared to the Atlantic. At face value, a stagnant deep ocean may explain the buildup of respired carbon reservoirs, resulting in a comparable basin-scale carbon sequestration pattern observed in both the Pacific and Atlantic regions (Du et al., 2018; Skinner et al., 2015). However, this hypothesis, which posits an extended transit time for the remineralization of organic matter, is challenged by εNd records from the Pacific that indicate a shorter transit time for deep Pacific water masses during glacial periods (Hu and Piotrowski, 2018). While the debate remains unresolved regarding whether there existed vigorous deep Pacific circulation (Du et al., 2020), our model results highlight the mechanisms that may sustain greater carbon storage, even in the presence of a vigorous circulation mode. Moreover, if there were stronger deep Pacific circulation during the glacials, the mechanisms influencing carbon storage at the basin scale in the two main oceans would differ markedly. Our objective is to evaluate the processes occurring in the Pacific and the Atlantic, with the intention of carefully distinguishing the similarities and differences present between them, as shown by the conceptual diagram in Fig. 7.

Our findings support the primary control of the SO in glacial marine carbon sequestration, aligning with the majority of existing literature (Du et al., 2022; Khatiwala et al., 2019; Marzocchi and Jansen, 2019; Sigman et al., 2021). This is robust for both the Pacific and the Atlantic. The cooling of the SO during glacial periods leads to a significant increase in sea ice and a change (or more likely, a decrease) in surface transit time, which impedes ocean outgassing, sequesters carbon in the ocean, enhances air-sea disequilibrium, and ultimately results in a reduction of pCO_2^{atm} levels. Crucially, our simulations demonstrate that in both basins, the cooling of the SO, along with increased brine rejection, could serve to strengthen the density structure within the deep ocean. In the absence of SO cooling, significant changes in the SSTs of the Northern Hemisphere may destabilize the deep stratification, resulting in carbon release from the deep ocean (as in IG_Gap).

Alternatively, we infer distinct modes of carbon sequestration in the NA and NP, which are closely connected to the regional oceanographic configuration. In the subpolar NA, the newly formed NADW exhibits a low surface pCO_2, acts as a carbon sink from the atmosphere, and continues to transport carbon into the ocean interior. During glacial periods, significant cooling and an increased meridional SST gradient would improve the efficiency of the solubility pump (Du et al., 2022; Yu et al., 2019), leading to greater carbon uptake in the NA. This signifies the glacial Atlantic carbon sequestration operating in an active carbon sequestration mode (Fig. 7, on the right), namely, to absorb CO_2 into the ocean steadily. In contrast, the subpolar NP features a lack of large-scale deep water formation (Marshall and Speer, 2012). During glacial periods, the reinforced stratification significantly reduces diapycnal mixing, preventing carbon from reaching the surface from the deep ocean. Consequently, we infer a passive carbon retention mode for the Pacific (Fig. 7, on the left). The 'passive' mode here refers to our results that the excess carbon is stored in the deep ocean because of less outgassing driven by the disequilibrium in the SO region and the halocline located in the subpolar NP.

5. Concluding remarks

In summary, by integrating the modeling datasets with prior records, we deduce that distinct carbon sequestration patterns could have operated in the glacial Pacific and Atlantic regions. Crucially, we validate that it is possible for the glacial Pacific to hold a large carbon pool even if more vigorous deep circulation was present. The dominance of SO-driven stratification in the deep ocean is combined with regional oceanographic configurations in determining the deep-ocean carbon

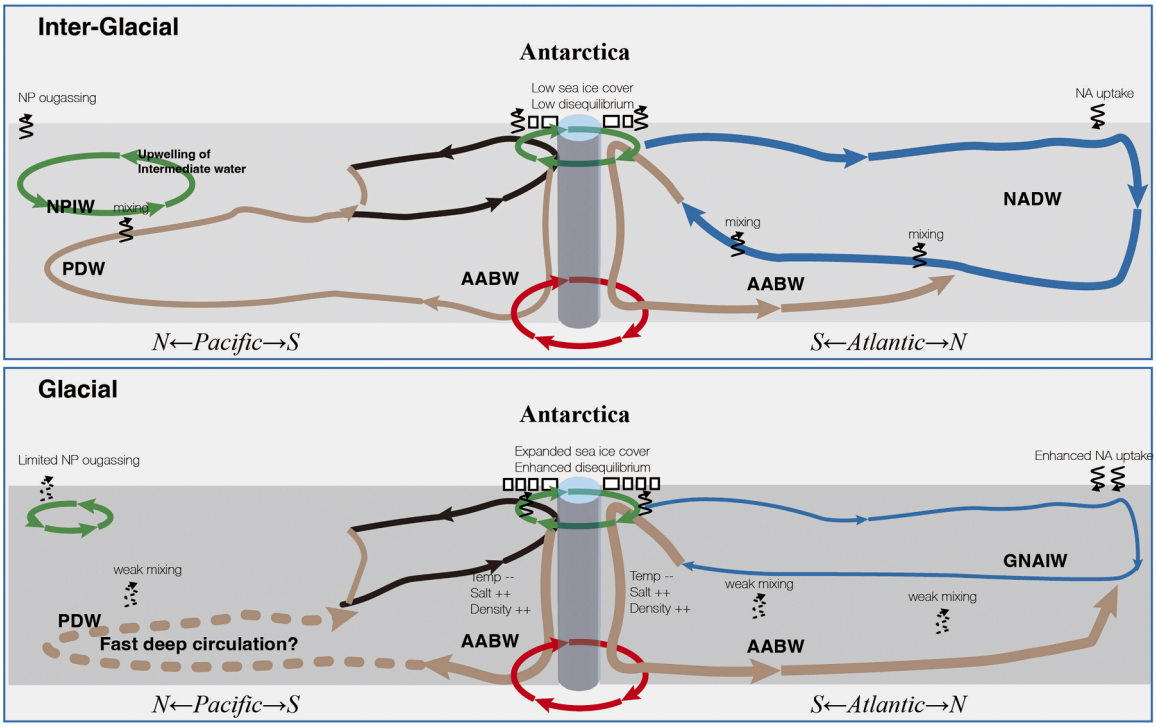


Fig. 7. A schematic diagram comparing the carbon dynamics operating in the Atlantic (right) and Pacific (left). The grayish shading in the two subplots indicates an overall increase in the oceanic carbon storage. The colored arrows refer to the general pattern of the thermohaline circulation. In the Atlantic basin, the NADW shoaled to the Glacial North Atlantic Intermediate Water, from the IG to G conditions. On the other hand, the deep Pacific may have exhibited a fast deep circulation, as indicated by εNd records. While the SO plays a pivotal role in stabilizing the stratification in the ocean interior and enhancing ocean-atmospheric disequilibrium, NA and NP both facilitate glacial carbon storage by enhancing surface uptake and restricting surface outgassing, respectively. The oceanographic configurations in the two basins therefore largely determine the different carbon sequestration patterns in the two oceans.

storage and sea surface carbon exchange. Within the framework of the thermohaline circulation, the glacial NA effectively absorbs atmospheric CO_2, while the surface SO and surface NP collaborate to inhibit oceanic outgassing.

CRediT authorship contribution statement

Jinlong Du: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Jun Tian:** Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2025.119344.

Data availability

All model outputs presented in this study are accessible to the public. Results of IG_ctl and G_ctl were previously published and can be found at https://doi.pangaea.de/10.1594/PANGAEA.935084. Results of IG_Gap and IG_Gsap can be found at https://zenodo.org/records/15069538.

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